SOFTWARE ARCHITECTURE & REQUIREMENTS SPECIFICATION

Interactive Elevation Takeoff & Estimation Tool

Client-Side Spatial Measurement, Quantity Takeoff & Dynamic Pricing Engine

1. Executive Summary & Core Objective

The objective of this project is to develop a lightweight, browser-based digital takeoff and estimation tool to replace manual Photoshop pixel measurements and disconnected spreadsheet workflows.

- **Core Purpose:** Enable estimators and project managers to visually calibrate scale, trace architectural components (e.g., lower wood panels, mirrored glass, decorative mouldings, brass appliques), auto-calculate surface areas and counts, and generate instantaneous bill of quantities (BOQ) with dynamic pricing.
- **Architecture Principle:** 100% client-side execution with zero recurring hosting costs, capable of being deployed on static platforms (Vercel, Cloudflare Pages) or self-hosted Nginx environments, serving seamlessly as an independent portfolio project.

2. Technical Stack & System Architecture

Layer / Domain	Recommended Technology	Purpose & Architectural Justification
Frontend Framework	**React (or Vue 3) + Vite**	Reactive state management for live pricing calculation updates; fast client-side rendering with modular component structure.
UI & Styling	**Tailwind CSS + Lucide Icons**	Clean, minimalist, responsive layout with floating toolbars, layer sidebars, and high-contrast control palettes.
Canvas Engine	**Fabric.js (or Konva.js)**	Hardware-accelerated HTML5 Canvas management handling viewport pan/zoom, shape drawing, polygon transforms, hit testing, and geometric grouping.
Pricing Database	**Google Apps Script (Web App)**	Exposes a master rate card from Google Sheets as a structured JSON endpoint, eliminating traditional SQL/ backend server management.
Local Persistence	**IndexedDB (Dexie.js / idb-keyval)**	Provides large local browser storage capacity to autosave active drawings, vector layers, and full Base64 background images seamlessly.
Export Engines	**SheetJS (*xlsx*) + Canvas / jsPDF**	Client-side compilation of structured Excel spreadsheets and high-resolution visual audit drawings with stamped dimensions and labels.
Hosting & CI/CD	**Vercel / Cloudflare Pages**	Zero-cost global static hosting with automatic Git deployments, free SSL certificates, and custom subdomain integration.

3. Comprehensive Feature Specifications

A. Image Ingestion & Spatial Calibration

- **Multi-Format Ingestion:** Drag-and-drop loading for PNG, JPG, WebP, and single-page raster/vector PDF drawings.
- **4-Point Perspective Warp (Keystone Correction):** Interactive quad-warp tool to square and flatten non-orthographic elevation photos before measurement.
- **Two-Point Scale Calibration:** User draws a reference vector across a known drawing benchmark (e.g., total wall height = *3200 mm*); system computes the global pixel-to-meter ratio (*px/m*).
- **Infinite Viewport Control:** Smooth hardware-accelerated pan (via mouse drag / spacebar) and continuous zoom (scroll wheel / pinch).

B. Annotation & Measurement Toolset

- **Area Tool (m^2):** Rectangles and multi-point freeform polygons for wall claddings, burl veneer panels, and glass inserts.
- **Count Tool (Points / Each):** Marker placement for discrete repetitive components (carved columns, pilasters, brass rosettes, lighting sconces).
- **Linear Tool (m):** Multi-segment polylines for top cornices, carved architraves, baseboards, and brass edge trims.
- **Deduction / Cutout Mode:** Geometric subtraction allowing inner openings (e.g., central portrait medallions or display niches) to be deducted automatically from parent surface areas.
- **Axis Locking & Duplication:** Hold ***Shift*** for strict ***0°, 45°, 90°*** constraints; ***Ctrl+D*** / ***Alt+Drag*** to instantly duplicate modular panel geometries.

C. Live Financial Estimation & Rate Modeling

- **Visual Material Categorization:** Layer-specific color tags (e.g., Gold = Veneer, Blue = Mirror, Bronze = Brass) rendered at *20%–30%* opacity with crisp border strokes.
- **Wastage & Cutting Factor:** Configurable percentage markup (*5%–20%*) applied per material tier to factor cutting losses.
- **Multi-Currency & Margin Sliders:** Dual currency exchange rate inputs with dynamic global markup multipliers.
- **Real-Time Header Summary:** Sticky HUD displaying **Grand Total Cost**, **Total Surface Area (m^2)**, and **Average Price per m^2** updated with each canvas interaction.

4. Data Storage & Lifecycle Strategy

```
[ User Canvas Interaction ]
        |
        ├──▶ Continuous Autosave ───▶ [ IndexedDB Storage ]
        |                             • Retains Base64 image + vector coordinates
        |                             • Instant recovery upon browser refresh/crash
        |
        ├──▶ "Save Project" ────────▶ [ Downloadable .json File ]
        |                             • Standalone portable project bundle
        |                             • Includes embedded image, scale, and metadata
        |
```

```
    └──▶ "Export Estimation" ────▶ [ SheetJS (.xlsx) / Apps Script (.doPost) ]
                                   • Formatted BOQ spreadsheet download
                                   • Direct row-by-row sync to master Google Sheet
```

- **Autosave:** Every point addition, rectangle drag, or rate adjustment commits instantly to local IndexedDB.
- **Project Portability:** Projects export as structured ***.json*** files with self-contained Base64 assets, allowing team sharing without broken image links.
- **Master Rate Sync:** On launch, the web app queries the Google Apps Script endpoint (***doGet***) to pull active unit rates, with fallback to local cache when offline.

5. Critical Implementation Guidelines & Watchouts

- **Normalized Coordinate System:** Canvas vector coordinates must be stored relative to native image dimensions rather than CSS display viewport bounds to prevent scaling errors across different monitor sizes.
- **High-Resolution Canvas Optimization:** For high-resolution files (>15MB), draw bounding boxes onto an overlay canvas while utilizing offscreen buffers for the background image to maintain a solid 60 FPS.
- **Visual Ghosting Key:** Bind a keyboard shortcut (e.g., holding ***Tab*** or ***Space***) to drop layer opacity to ***0%*** for instant inspection of underlying millwork lines.
- **CORS Header Handling:** Ensure the Google Apps Script endpoint returns explicit ***Access-Control-Allow-Origin: **** headers to permit cross-origin asynchronous fetch requests.